



HOLY SPIRIT UNIVERSITY OF KASLIK
SCHOOL OF ENGINEERING

HRI REPORT

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Introduction

This project presents the development of an interactive system using the Pepper humanoid robot as an AI teaching assistant in a classroom environment. The goal is to enhance student engagement through speech interaction, tablet-based interfaces, animations, and smart communication features. Pepper is designed to present educational content, interact with users, display information on its tablet, and collect attendance and survey responses.

The system integrates Pepper's NAOqi services with external laptop processing to extend its capabilities, including image-based fun filters and interactive activities. This project demonstrates how social robots can support modern learning environments by making human-robot interaction more natural, engaging, and effective.

Project proposal and motivation

Description:

Pepper will be acting as a friendly teaching assistant in a Technology class at USEK. The robot is designed to support the professor by managing visual aids, engaging with students through humor, and facilitating a high-tech learning environment.

Motivation:

This project aims to elevate the educational experience at USEK by integrating the Pepper robot and Gemini AI as an interactive teaching assistant. By blending academic support with engagement tools, such as humor-based icebreakers and real-time student scanning, the goal is to significantly boost classroom participation.

Activities and tools:

- **Robot & Development:** Pepper Robot controlled via its corresponding IDE.
- **Attendance & Security:** Python-based Face-Recognition for automated attendance taking and database registration.
- **Interactive Presentation:** Pepper assists the teacher by displaying specific images on its tablet.
- **Advanced Q&A (Powered by Gemini):** Once the presentation ends, Pepper scans the students to identify those with questions and approaches the student who raises their hand.

Using Gemini integration, Pepper provides intelligent, real-time answers while approaching the student who raised their hand and maintaining a safe interaction distance during the conversation.

- **Interactive Student-Based Quiz:** After answering the students' questions, Pepper launches a short quiz made of 5 questions generated from the students' Q&A during the session. Using its tablet and voice interaction, Pepper presents the questions one by one, allowing students to participate and test their understanding in a fun and interactive way.
- **AI Picture Filters & Interactive Experience:** After the Q&A session, Pepper captures students' photos and applies fun AI-generated filters to them. While the image is being processed, Pepper plays a waiting song to entertain the students. Once completed, a QR code containing the generated image is displayed on Pepper's tablet so students can download it directly to their phones. A "Try Again" button is also provided, allowing multiple students to enjoy the experience.

- **Attendance and Survey Module:** At the end of the session, Pepper displays an Attendance and Survey interface on its tablet, allowing students to register their presence and provide feedback about the presentation. The attendance data is automatically collected and saved in an Excel file, which is then sent via email to the instructor for easy tracking and record keeping. This feature helps organize participation monitoring while also gathering valuable student feedback to evaluate the session.

Project use cases (4)

Use case ID/Name	1 – Assist Course Introduction
Objective or goal	The objective of this use case is to initiate the start of the program, by having pepper assist the doctor in their course introduction about AI, being by adding fun comments or illustrations on the tablet for better understanding.
Actors	Doctor Maria, the doctor tasked with teaching the AI course. Pepper, the doctor assistant, helping Dr Maria.
Preconditions	The system is started from an Idle state.
Postconditions	Pepper waits for the Q&A interaction.
Action Sequence (AS)	1- Doctor: Initializes the conversation. 2- Pepper: Introduces itself + shows image. 3- Doctor: Starts explaining the course. 4- Pepper: Continues and Explain the Doctor's explanations by examples and images. 5- Pepper: Ends the course introduction.
Storyboard	USE CASE 1 – ASSIST COURSE INTRODUCTION – STORYBOARD (1/2) 1 Doctor: Initializes the conversation Introduction to Artificial Intelligence Exploring the Future of Technology Today Good morning I am doctor maria. Doctor Maria greets the class and initializes the conversation. 2 Pepper: Introduces itself + shows image Introduction to Artificial Intelligence Hello everyone, I am Pepper, your AI Teaching Assistant. Don't worry... I won't replace your professor. Yet. *pepper winks* Pepper introduces itself to the class with a friendly greeting and a joke. 3 Doctor: Starts explaining the course Introduction to Artificial Intelligence Exploring the Future of Technology Today Doctor Maria begins explaining the topic of Artificial Intelligence. 4 Pepper: Continues and Explain the Doctor's explanations What is AI? • AI allows machines to think, learn, and make decisions. • AI systems use algorithms and data to detect patterns and improve performance over time. Technically speaking, AI systems use algorithms and data to detect patterns and improve performance over time. Pepper supports the doctor's explanation with additional details and examples. 5 Pepper: Shows illustrations on tablet How AI Works Data → Processing → Learning → Decision Pepper displays relevant illustrations and visuals on its tablet to make the concepts easier to understand. 6 Pepper: Ends the course introduction Introduction to Artificial Intelligence Exploring the Future of Technology Today I am ready Pepper ends the course introduction and prepares the class for the next part.
Requirements	Functional: 1- Voice Recognition: Pepper can understand and talk to the doctor. 2- Tablet Display: Pepper can display the illustrations and images.

Use case ID/Name	1 – Assist Course Introduction
	Nonfunctional: 1- Low Delay: Pepper should be fast with the responses to not break the flow of the conversation and presentation. 2- Reliability: Pepper should have a stable connection and make minimal errors to not break the illusion.

Use case ID/Name	2 – Answer Student’s Questions
Objective or goal	The objective of this use case is for pepper to talk to the students and answer their questions using the AI integration.
Actors	Students, which would ask pepper questions about topics related to the lesson. Pepper, would answer the question based on the connected AI.
Preconditions	Pepper initializes this after finishing the lesson introduction.
Postconditions	Back to Idle State.
Action Sequence (AS)	1. Pepper: Initiates the Q&A session. 2. Student: Raises their hand to request interaction. 3. Pepper: Detects the raised hand and approaches the student. 4. Student: Asks a question related to the AI course. 5. Pepper: Processes the question and provides an answer using the connected AI system.

Use case ID/Name	2 – Answer Student’s Questions
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Storyboard

USE CASE 2 – ANSWER STUDENT’S QUESTIONS – STORYBOARD

1

Pepper: Initiates the Q&A session.

Pepper invites the class to start the Q&A session.

2

Student: Raises their hand to request interaction.

A student raises their hand to ask a question.

3

Pepper: Detects the raised hand and approaches the student.

Pepper detects the raised hand and moves closer to the student.

4

Student: Asks a question related to the AI course.

The student asks a question related to the AI course.

5

Pepper: Processes the question using the connected AI system.

Pepper sends the question to the AI system and analyzes the information.

6

Pepper: Provides an answer using the connected AI system.

Pepper delivers a clear and accurate answer based on the AI system.

Requirements

- Functional:

1. **AI connection:** Pepper should be connected to an AI so that it would be able to answer the questions.

2. **Voice Recognition:** Pepper can understand and talk to the doctor and students.

3. **Managed Response:** The AI should not answer any question; it should only answer questions related to the course.

Nonfunctional:

4. **Reliability:** Pepper should have a stable connection and make minimal errors to not break the illusion.

Use case ID/Name	3 – Conduct Pop Quiz
Objective or goal	The objective of this use case is to allow Pepper to start a short pop quiz after the course introduction or Q&A session, ask students course-related questions, collect their answers, and record their performance to help the doctor know what topics students understood or found difficult.
Actors	Students, who answer the pop quiz questions. Pepper, which asks the questions, receives the answers, and records the results.
Preconditions	Q&A session has ended.
Postconditions	Pepper records the students' answers and scores. Pepper identifies the topics that students found difficult. The system returns to Idle State.
Action Sequence (AS)	1. Pepper announces the beginning of the pop quiz. 2. Pepper displays the quiz instructions on the tablet. 3. Pepper asks a course-related question. 4. Students answer using Pepper's tablet or by voice. 5. Pepper checks the students' answers. 6. Pepper shows student's grade. 8. Pepper ends the pop quiz.
Storyboard	USE CASE 3 – CONDUCT POP QUIZ – STORYBOARD 1 Pepper announces the beginning of the pop quiz. Let's start a short pop quiz to test what you learned! POP QUIZ Pepper announces that the pop quiz is starting. 2 Pepper displays the question and reads it. QUESTION 1: What is AI mainly based on? A) Hardware B) Data and Algorithms C) Electricity D) Human Emotions Pepper shows the question on its tablet and reads it to the student. 3 Students answer using Pepper's tablet or by voice. QUESTION 1: What is AI mainly based on? A) Hardware B) Data and Algorithms C) Electricity D) Human Emotions I think the answer is B. Students select or say their answer using the tablet or voice. 4 Pepper checks the students' answers. Checking answer... Pepper processes the answer and verifies it using the AI system. 5 Pepper shows student's grade and gives feedback. POP QUIZ RESULT Score: 4/5 Great job! You scored 4 out of 5! Great job! 6 Pepper ends the pop quiz. Pop quiz complete! Well done everyone! See you in the next session! Pepper thanks the student and ends the pop quiz.
Requirements	Functional: 1. Quiz Display: Pepper should display quiz questions and possible answers on the tablet.

Use case ID/Name	3 – Conduct Pop Quiz
	2. Answer Collection: Pepper should allow students to answer using the tablet or voice recognition. 3. Answer Checking: Pepper should compare student answers with the correct answers. Nonfunctional: 4. Performance: Pepper should respond quickly after each answer.

Use case ID/Name	4 – Apply Fun Image Filters
Objective or goal	The objective of this use case is to allow Pepper to take a picture of the students, apply fun AI-generated filters to the image, and display the final edited result on Pepper's tablet for an engaging classroom experience.
Actors	Students, who choose a filter and pose for the picture. Pepper, which takes the picture, sends it to the AI image filter system, and displays the edited result.
Preconditions	Pepper is connected to the image filtering system. Pepper's camera is working.
Postconditions	Pepper continues to the next interaction collecting attendance and survey.
Action Sequence (AS)	1. Pepper starts the fun image filter activity. 2. Pepper asks the students to choose a filter from the tablet. 3. Students select a filter, such as sunglasses, hat, or funny hair. 4. Pepper takes a picture of the students using its camera. 5. Pepper sends the captured image to the connected AI image filter system. 6. The AI system applies the selected filter to the image. 7. Pepper receives the edited image. 8. Pepper displays the final filtered image on the tablet. 9. Pepper asks if the students want to try again or continue.

Use case ID/Name	4 – Apply Fun Image Filters
Storyboard	USE CASE 4 – APPLY FUN IMAGE FILTERS – STORYBOARD 1 Pepper starts the fun image filter activity. Let's have some fun! We'll take a photo and apply cool AI filters! Pepper introduces the fun image filter activity to the students. 2 Pepper asks the students to choose a filter from the tablet. Please choose a filter you'd like to use. Pepper displays the filter options on its tablet and asks the students to choose. 3 Pepper takes a picture of the students using its tablet. Say cheese! Pepper captures a photo of the students using its tablet. 4 Pepper sends the photo to the AI system and applies the selected filter. Applying filter... Please wait while I work my magic! Pepper sends the photo to the AI system and applies the selected filter. 5 Pepper displays the final filtered image. Here's your filtered photo! Pepper shows the edited image to the students on its tablet. 6 Students scan the QR code to get their photo. Scan this QR code on your phone to get your photo! Students scan the QR code with their phones. The final photo will be available on their device.
Requirements	Functional: Face recognition: Pepper should be connected to a program that would let it be able to recognize the student faces. Camera: Pepper is equipped with a camera that would be high quality enough for the Facial recognition to work. Movement: Pepper should be able to approach the students when asking questions. Nonfunctional: Reliability: Pepper should have a stable connection and make minimal errors to not break the illusion.

Use case ID/Name	5 – Record Attendance and Collect Session Survey
Objective or goal	The objective of this use case is to allow Pepper to record students' attendance using face recognition and collect their feedback through a survey displayed on Pepper's tablet.
Actors	Students, whose attendance is recorded and who complete the session survey. Pepper, which scans the students' faces, records attendance, displays the survey QR code, and guides the

Use case ID/Name	5 – Record Attendance and Collect Session Survey
	students. Doctor Maria, who can later review the attendance sheet and survey responses.
Preconditions	The image filter activity has ended.
Postconditions	Back to Idle State
Action Sequence (AS)	1. Pepper announces the attendance and survey phase. 2. Pepper records the recognized students as present in the attendance database. 3. Pepper displays the survey option or QR code on the tablet. 4. Students scan the QR code or open the survey from Pepper's tablet. 5. Students submit their session ratings. 6. Pepper confirms that the attendance and survey process is completed. 7. Pepper thanks the students and returns to Idle State.
Storyboard	USE CASE 5 – RECORD ATTENDANCE AND COLLECT SESSION SURVEY – STORYBOARD 1 Pepper introduces the attendance and survey phase. Now we'll record attendance and then complete a short session survey! Pepper informs the students that it will now record attendance and collect their feedback. 2 Students click on "Survey" to enter the survey part. Students choose "Survey" on Pepper's tablet to access the session survey. 3 Pepper displays a QR code for the survey. Please scan the QR code to take the survey. A QR code is displayed on Pepper's tablet. Students scan it using their phones to open the survey. 4 Students scan the QR code and submit their session rating. Students scan the QR code, rate the session with stars, add any comments, and submit. 5 Pepper records attendance and sends the attendance list to the doctor. Attendance list has been sent to your email. To: doctormaria@clinic.com Sent successfully! Pepper uses face recognition to record attendance and sends the attendance list to the doctor's email as an Excel file. 6 Pepper shows a thank you message as an outro. Thank you! Pepper thanks the students for their participation and concludes the session.
Requirements	Functional: 1. Attendance Recording: Pepper should record recognized students as present in the database or attendance sheet. 2. Survey Display: Pepper should display the survey QR code or survey page on its tablet. 3. Survey Collection: Students should be able to submit their ratings . Nonfunctional: 4. Reliability: Pepper should record attendance accurately and avoid missing students.

Personas

Personas related to designing a Robot Teacher Assistant for Students in a GIN Class

 "I want to spread my knowledge"		
Demographics	Name	Dr Maria
	Age	32
	Degree	PHD in Computer Engineering
	Number of Classes	7
	Office	H308
Motivation	Hobbies	Music And Photography
	Sports	Badminton
	Work	She is disciplined, motivated and competitive.
	Personal challenges	Manages both work and side projects so she has difficulty in managing her time, which bottlenecks her performance.
Social Environment	Background	Middle-class family from Jounieh. Father was a local school principal; mother was a librarian. She values the prestige and community feel of USEK.
	Parents	Married with two children.



“I want to learn new things.”

Demographics	Name	Charbel
	Age	21
	Year of study	4
	Major	Computer Engineering
	Class	GIN
Motivation	Hobbies	Travelling
	Sports	Football
	School	Highly motivated when doing hands on projects and work, but loses focus when reading slides.
	Personal challenges	Having difficulty understanding class material because he gets easily distracted or finds no real-life use to it.
Social Environment	Background	Father has very long working hours while the mother stays at home to cook and care for the children.
	Friends	Prefers dynamic studying with his friends in order to have fun and keep it going.



“I feel like this isn’t for me.”

Demographics	Name	Kevin
	Age	23
	Year of study	5
	Major	Computer engineering
	Class	GIN425
Motivation	Hobbies	Stock trading and going out
	Sports	Basketball
	School	He wants to graduate and prove he can finish the degree, but he often diverts his attention to other interests (like building stock trading bots) when the core engineering theory gets too dense.
	Personal challenges	Always feels like he will not be able to make it in life due to the competitiveness of the real world.
Social Environment	Background	Upper class family from Beirut. His parents own multiple lands and companies, and he is set for years. He is studying this major just for the sake of it.
	Friends	Has a lot of friends, but when it comes to studying it’s mostly alone that he does it.



“I thrive to be an overachiever.”

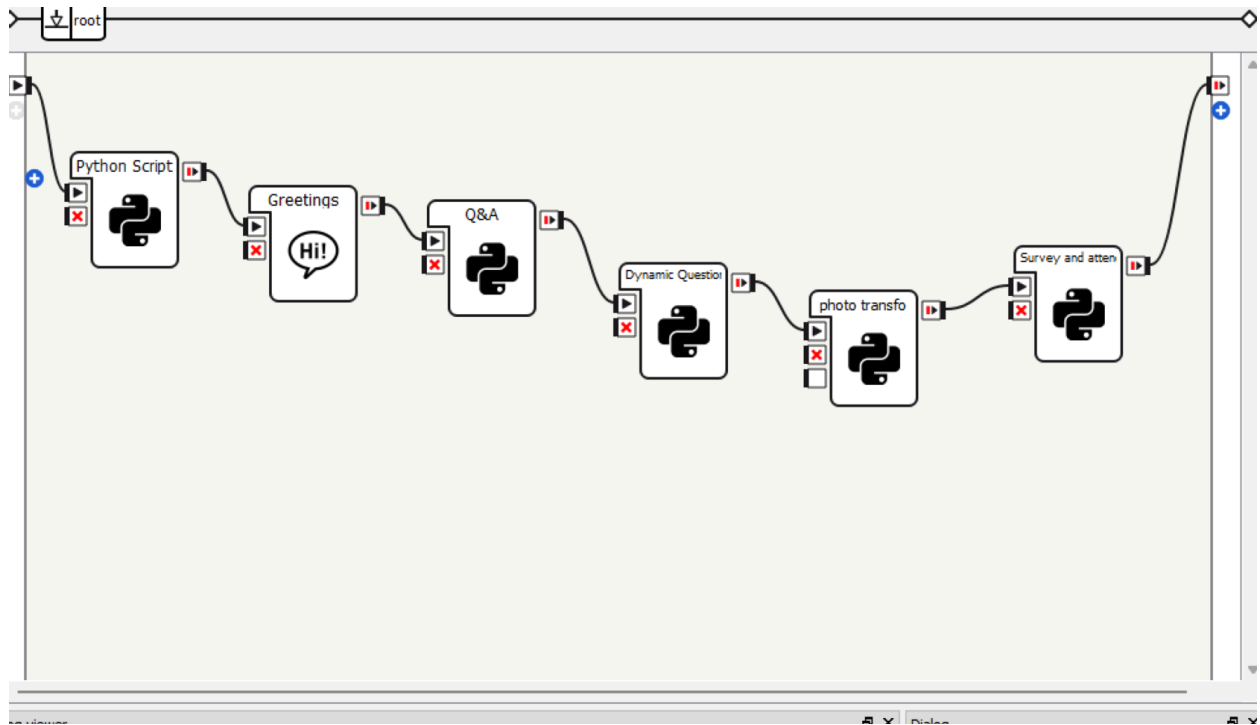
Demographics	Name	Tatiana
	Age	22
	Year of study	4
	Major	Computer engineering
	Class	GIN425
Motivation	Hobbies	Music, Reading, Coding
	Sports	Running daily
	Part-time Job	Tatiana works as a student assistant in the university robotics lab. She helps maintain equipment and assists students working on robotics and AI projects.
	Personal challenges	Sometimes feels stressed when managing multiple assignments and responsibilities at the same time, especially during busy academic periods.
Social Environment	Background	She grew up fascinated by technology and computers. This curiosity pushed her toward engineering.
	Friends	Tatiana has a small but supportive social circle made mostly of classmates. They often study together and motivate each other to succeed academically.



“No pain, no gain.”

Demographics	Name	Joseph
	Age	21
	Year of study	3
	Major	Computer engineering
	Class	GIN425
Motivation	Hobbies	Playing guitar and video games
	Sports	Gym
	School	Engineering student, chose this major because he enjoys the concepts a lot and sees where the world is heading, uses it as motivation.
	Personal challenges	Procrastinates a lot and finds time management to be a tough thing, so he ends up late at night studying last minute.
Social Environment	Background	Middle class family from Jounieh. Only child of a stay at home mom and working dad, he looks up to them a lot in his daily life.
	Friends	A small circle he always hangs out with and thrives to always achieve better academically with them.

Implementation



Description of the Interaction

The Choregraphe behavior is organized as a complete Pepper classroom flow. It starts with a Python Script box, then moves to the **Greetings** dialog, followed by the **Q&A** module, the **Dynamic Question / Pop Quiz** module, the **Photo Transformation** module, and finally the **Survey and Attendance** module.

Pepper first introduces the AI lesson, then answers students' questions using AI, generates quiz questions, applies fun image filters, and ends by showing attendance and survey options on the tablet.

HTML / CSS / JS Templates Used

Main Menu

- index.html
- index.css
- index.js

Used for the final tablet menu with two buttons:

- **Attendance**
- **Survey**

The `index.js` file also raises the `NextPressed` event when the Next button is clicked.

Attendance Page

- `attendance.html`
- `attendance.css`
- `attendance.js`

Used to display recognized students, show the attendance list, request a password, and send the attendance Excel file by email.

Survey Page

- `survey.html`
- `survey.css`
- `survey.js`

Used to display the survey QR code on Pepper's tablet and allow students to scan it.

Quiz Page

- `confirmation.html`
- `confirmation.css`
- `confirmation.js`
- `questions.json`

Used for the pop quiz. Pepper displays each question, reads it aloud, collects answers, checks the correct answer, and calculates the score.

Chat Display

- `chat.html`
- `chat.css`
- `chat.js`

Used to display the Q&A conversation on Pepper's tablet, showing student questions and Pepper's AI answers.

Image Filter Pages

Added inside `combine.py` as HTML templates:

- `CHOICE_HTML`
- `LOADING_HTML`
- `RESULT_OPTIONS_HTML`

Used for:

- choosing a filter: sunglasses, hat, funny hair, cartoon
- showing a loading screen with waiting music
- displaying QR code to download the generated image
- choosing Try Again or Next

Backend Functions Added / Modified

`combine.py`

This file combines the main backend features:

- Starts Flask server on port 5000
- Starts attendance/email server on port 5001
- Detects laptop IP automatically
- Generates `config.js`
- Connects to MySQL attendance database
- Creates attendance Excel file
- Sends the Excel file by Gmail
- Sends images to Gemini for AI filters
- Generates QR code for the edited image
- Loads quiz questions from `questions.json`

Challenges Faced by Each Member of the Group

Members Tatiana & Maria – Attendance Excel, Email System & AI Picture Filters

- Images sometimes showed only on the web and not on Pepper's tablet.
- AI filters and effects were not always applied correctly.
- Some images did not display properly or did not appear at all.
- Synchronization issues occurred between Pepper's animations and voice.
- Attendance emails sometimes failed to send due to connection and authentication issues.

Member Charbel– Gemini Q&A Integration

- Faced difficulties integrating Gemini AI with Pepper in real time.
- Hand-raised detection and room mapping were not always accurate.
- Pepper movement and navigation sometimes caused positioning issues.
- QR code generation and display occasionally failed.
- Synchronizing AI responses between screen display and Pepper's voice was challenging.

Member Kevin – Student-Based Pop Quiz

- Faced difficulties generating quiz questions correctly from the JSON data.
- Managing score calculation and updating results dynamically was challenging.
- Synchronizing question display with Pepper's voice reading required timing adjustments.
- Ensuring smooth interaction between students and the quiz system was difficult at times.

Member Joseph – Pepper Gesture Signs

- Synchronizing the goodbye image display with Pepper's waving animation and voice.

- Faced timing issues between the tablet display and movement execution.
- Ensuring smooth and natural waving gestures without movement errors was challenging.

Conclusion

This project demonstrated how the **Pepper humanoid robot** can be used as an interactive teaching assistant to enhance classroom engagement through speech interaction, tablet interfaces, AI-powered question answering, face recognition attendance, pop quizzes, picture filters, and automated survey collection. By integrating Pepper's NAOqi services with external tools such as Gemini AI, Gmail-based reporting, and image-processing modules, the system provided a complete smart classroom support solution. Despite technical challenges related to camera usage, module communication, and system synchronization, the project successfully showed how social robots can improve participation, interaction, and learning experiences in modern educational environments.